

99.500 Applied Mathematics for Engineering

Homework 1

February 4, 2026

1. A semi-infinite slab is initially at a uniform temperature T_i . At time $t = 0$, the temperature at the surface begins to increase linearly with time, following the relation:

$$T(x = 0, t) = T_i + At$$

where A is a constant. Within the slab, heat transfer occurs solely by conduction, governed by Fourier's law.

- (a) Using the shell balance technique, derive the governing equation for the non-steady-state temperature distribution $T(x, t)$.
 - (b) Determine the appropriate boundary and initial conditions for this problem.
2. A water stream with a constant volumetric flow rate Q is fed into a pipe of radius R with a permeable wall, as illustrated below. The pipe is embedded in a porous medium that extends infinitely in the radial direction but has a finite height L . The water contains a dissolved reagent at an initial concentration C_F which undergoes a first-order reaction on the surface of the solid particles within the porous medium. The reaction rate is given by $r_s = kC \left(\frac{\text{mole}}{\text{cm}^2\text{s}} \right)$. The molar flux of the reagent N is expressed as

$$N = Cv - D \frac{dC}{dr}$$

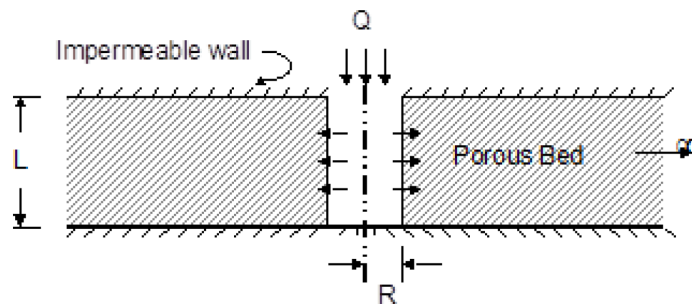
where v is the superficial velocity and D is the effective diffusivity.

- (a) Using the shell balance technique, show that the steady-state concentration distribution is governed by the equation:

$$r^2 \frac{d^2 C}{dr^2} + r \left(1 - \frac{rv}{D} \right) \frac{dC}{dr} - \frac{ak}{D} r^2 C = 0$$

where a is the specific interfacial area of the porous solid $\left(\frac{\text{cm}^2}{\text{cm}^3} \right)$. Clearly state any assumptions made in the derivation. [Hint: For dilute solutions, mass continuity requires $\frac{d}{dr}(rv) = 0$].

- (b) Identify the appropriate boundary conditions for the governing equation. Hint: Consider the two physical boundaries in the system.
- (c) Nondimensionalize the governing equation and its boundary conditions.



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Homework 2

February 25, 2026

1. A polymer film of thickness d is coated onto a metal substrate. The film's surface at $x = 0$ is exposed to air and irradiated by an ultraviolet (UV) source. This irradiation induces a photochemical cleavage reaction, producing monomers at a local volumetric rate of r_m with unit $\frac{\text{mol}}{\text{cm}^3\text{s}}$. The reaction can be represented as: $(M)_n \rightarrow M + (M)_{n-1}$, where $(M)_n$ is the polymer, M is the monomer, and $(M)_{n-1}$ is the polymer with one fewer monomer unit. The monomer is soluble in the polymer and, since the substrate is impermeable, it can only diffuse out of the film through the $x = 0$ surface. The depolymerization kinetics follow zero-order behavior with respect to the dissolved monomer concentration, but the reaction rate decreases with depth due to the absorption of UV radiation, following an exponential attenuation law as

$$r_m = r_{m_0} e^{-\lambda x} \quad (1)$$

where r_{m_0} is the production rate at the surface $x = 0$ and λ is the absorption coefficient that characterizes how strongly the polymer absorbs the UV light. Assuming Fickian diffusion governs the transport of the dissolved monomer and that a pseudo-steady-state condition applies, the governing equation and boundary conditions for the monomer concentration are given as

$$D \frac{d^2 C}{dx^2} = -r_m \quad (2)$$

$$C(0) = 0 \text{ \& \> } \left. \frac{dC}{dx} \right|_{x=d} = 0 \quad (3)$$

Solve the above ODE to obtain an explicit expression for the monomer concentration.

2. A thin, disk-shaped platinum (Pt) particle with radius R in cm is positioned on a Zirconium dioxide (ZrO_2) surface of infinite extent, and the entire system is exposed to a dioxygen (O_2) atmosphere. The Pt selectively chemisorbs dioxygen at its perimeter, forming oxygen adatoms. The ZrO_2 surface itself does not adsorb oxygen. The oxygen adatoms generated at the particle perimeter diffuse along the Pt/ZrO_2 interface and react with oxygen vacancies in the ZrO_2 crystal lattice. This reaction follows first-order kinetics with respect to the oxygen adatom concentration. The diffusion of oxygen adatoms along the Pt/ZrO_2 interface is governed by Fickian diffusion, with a surface flux N_s with unit $\frac{\text{mol}}{\text{cm s}}$ as

$$N_s = -D_s \frac{dC_s}{dr} \quad (4)$$

where C_s is the surface concentration of adsorbed oxygen in $\frac{\text{mol}}{\text{cm}^2}$ and D_s is the surface diffusivity constant in $\frac{\text{cm}^2}{\text{s}}$. Assume that the number density of oxygen-vacancy sites at the Pt/ZrO_2 interface remains constant, meaning that the ZrO_2 acts as an infinite sink for oxygen. Additionally, suppose that the adsorbed oxygen at the disk perimeter is in equilibrium with the gas phase, leading to a known adsorbed oxygen concentration at radius R , denoted as C_{s_0} . The governing equation for the oxygen adatoms and its boundary conditions can be derived as

$$r^2 \frac{d^2 C_s}{dr^2} + r \frac{dC_s}{dr} - r^2 \frac{k}{D_s} C_s = 0 \quad (5)$$

$$C_s(R) = C_{s_0} \text{ \& \> } \left. \frac{dC_s}{dr} \right|_{r=0} = 0 \quad (6)$$

where k is the pseudo first-order rate constant for the reaction between oxygen adatom and oxygen vacancies.

(a) Solve the above ODE to obtain an explicit expression for $C_s(r)$. (ODE derivation is not required).

(b) The steady-state reaction rate of absorbed oxygen R_0 for the entire disk can be calculated by evaluating the flux at the surface, $R_0 = -2\pi RN|_{r=R}$. Calculate R_0 in terms of R .

(c) Sketch a plot of R_0 vs R and investigate the “small R ” and “large R ” behaviours. (Please include the slope and the asymptotic value of the plot)

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Homework 3 (Q1 is compulsory, followed by either Q2 or Q3)

March 24, 2026

1. A solid sphere, initially at a uniform temperature T_0 , is suddenly immersed in a constant-temperature bath maintained at T_∞ . It is assumed that the bath is not vigorously stirred, and the external heat-transfer resistance is negligible, implying that the surface temperature of the sphere remains at T_∞ .
 - (a) Determine the unsteady-state temperature profile within the sphere.
 - (b) Derive an expression for the sphere's average temperature as a function of time.
2. Use the separation of variables to find the solution $C(x, t)$ for the following problem:

$$\frac{\partial C}{\partial t} = \frac{1}{x} \frac{\partial}{\partial x} \left(x \frac{\partial C}{\partial x} \right) \quad (1)$$

$$C(x, 0) = 1 \quad (2)$$

$$\frac{\partial C}{\partial x} \Big|_{x=1} = 0 \quad (3)$$

$$\alpha \frac{\partial C}{\partial x} \Big|_{x=\beta} = C(\beta, t) \quad (4)$$

where α and β are constants.

3. Use the separation of variables to find the solution $C(x, t)$ for the following problem:

$$\frac{\partial C}{\partial t} = \frac{1}{x} \frac{\partial}{\partial x} \left(x \frac{\partial C}{\partial x} \right) - \delta C \quad (5)$$

$$C(x, 0) = 1 \quad (6)$$

$$\frac{\partial C}{\partial x} \Big|_{x=0} = 0 \quad (7)$$

$$C(1, t) = 0 \quad (8)$$

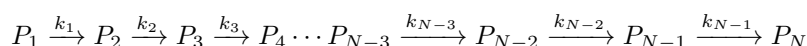
where δ is constant.

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Homework 4 (Q1 is compulsory, followed by either Q2 or Q3)

December 28, 2025

1. In a cutting-edge biotechnology lab, a specialized system is designed to produce a high-value therapeutic protein. This system involves a series of cellular processes in which raw biological materials undergo irreversible transformations to produce the final therapeutic product. The process occurs in $N - 1$ stages, each representing a different step in the cellular machinery, as shown below:



The production starts with an initial substrate, such as a nutrient or precursor protein, denoted as P_1 , which is introduced into the first stage of the cellular process. Within this stage, a reaction converts some of P_1 into an intermediate protein P_2 with rate constant k_1 . This intermediate then proceeds to the next stage, where it is further converted into P_3 with rate constant k_2 . The process continues through a series of stages until the final therapeutic protein P_N is produced with rate constant k_{N-1} . The concentration of P_1 is C_{initial} at time $t = 0$, while all the other proteins have 0 concentration.

The researchers are particularly interested in monitoring the concentration of each protein intermediate throughout the production process to ensure efficiency and optimize yields. To achieve this, use Laplace transform to derive a general formula for the concentration of each protein at any given time, and provide the formula for the concentrations of the first three proteins.

2. Use the separation of variables to find the solution $T(r, z)$ for the following problem:

$$\frac{\partial^2 T}{\partial z^2} + \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) = 0 \quad (1)$$

$$T(r, 0) = 0 \quad (2)$$

$$\frac{\partial T}{\partial z} \Big|_{z=\alpha} = 0 \quad (3)$$

$$\frac{\partial T}{\partial r} \Big|_{r=0} = 0 \quad (4)$$

$$\frac{\partial T}{\partial r} \Big|_{r=1} = f(z) \quad (5)$$

where α is constant. What is $T(r, z)$ for the case $f(z) = 1$?

3. Use the separation of variables to find the solution $T(r, z)$ for the following problem:

$$\frac{\partial^2 T}{\partial z^2} + \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) = 0 \quad (6)$$

$$T(r, 0) = T_0 \quad (7)$$

$$\frac{\partial T}{\partial z} \Big|_{z=L} = \begin{cases} \beta & r \leq r_1 \\ 0 & r_1 < r < r_2 \end{cases} \quad (8)$$

$$\frac{\partial T}{\partial r} \Big|_{r=0} = 0 \quad (9)$$

$$\frac{\partial T}{\partial r} \Big|_{r=r_2} = 0 \quad (10)$$

where β , r_1 and r_2 are constants.

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Homework 5

April 7, 2026

1. A semi-infinite slab is initially at a uniform temperature T_i . Starting at time $t = 0$, the surface temperature increases linearly with time, given by:

$$T(x = 0, t) = T_i + \alpha t \quad (1)$$

where α is a constant. Using the Laplace transform technique, derive an expression for the heat flux q at the surface of the slab as a function of time.

2. A solid cylinder is initially at a uniform temperature T_i . At time $t > 0$, the temperature at its outer surface where $r = r_o$ is suddenly changed to T_0 . Using the Laplace transform method, determine the temperature distribution $T(r, t)$ by solving the following heat conduction equation in cylindrical coordinates:

$$\frac{\partial T}{\partial t} = \beta \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) \quad (2)$$

where β is constant.